Implementation Roadmap

AU Roadmap Project — Adelaide University

Project Name: Adelaide University Program Roadmap
Course: Industry Research Project
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Group Code: C261W-4037

Team Members:
Jialao Liu
Yuanxin Huang
Yilin Peng
Vatsal Thanki

Academic Mentor: Wenhao Liang

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Revision History

Version	Date	Author	Summary of Revision
0.1	Mar. 2026	J. Liu, Y. Huang, Y. Peng	Initial project phase planning in proposal
0.5	Apr. 2026	J. Liu, Y. Huang, Y. Peng, V. Thanki	Development and implementation phase structure refined
0.8	May 2026	J. Liu, Y. Huang, Y. Peng, V. Thanki	Prototype completion and testing phase clarified
1.0	May 10, 2026	J. Liu, Y. Huang, Y. Peng, V. Thanki	Formal implementation roadmap updated to reflect actual project status and future direction

Table 1. Document revision history.

1. Introduction

The Adelaide University Program Roadmap project has moved from an initial proposal through to a working multi-role prototype supported by testing and documentation. This roadmap traces the completed path, summarises the current state, and identifies practical next steps for continued development.

The roadmap connects three layers: the original Project Proposal [1], the implemented prototype, and possible post-prototype extensions. It serves both a retrospective function—documenting what has been built and tested—and a forward-looking function—structuring remaining work into actionable phases. This dual perspective demonstrates not only delivery but also a managed, sustainable continuation path.

2. Roadmap Purpose and Scope

2.1 Purpose

The roadmap serves four purposes: summarise the current project stage; show how completed work aligns with the planned project structure; identify the major work phases that have been finished; and clarify dependencies, risks, and feasibility for any continuation beyond the prototype.

2.2 Scope

The document covers implementation progress across planning, design, development, testing, and documentation; future technical and functional extension priorities; and a practical sequence for post-prototype continuation. It functions as a high-level structured execution roadmap suitable for project reporting and future planning. It does not serve as a deployment runbook, a detailed sprint backlog, or a production delivery contract.

3. Project Overview

The Adelaide University Program Roadmap project was proposed as a web-based platform with two levels. Level 1 supports current students through roadmap, timetable, support, and related academic and employability information. Level 2 supports prospective students through public program exploration, outcomes, campus life, and application guidance. The project also developed an admin management panel to support structured content maintenance.

The overall implementation objective was to deliver a structured, feasible, and extensible system concept supported by documentation, testing, and role-based architecture, not merely a prototype interface. This approach aligns with project management best practice for phased delivery of complex systems [2].

4. Current Project Status

4.1 Overall Status

The project is at a working prototype stage. The system has moved beyond planning and design into a complete multi-role implementation with supporting technical and testing documentation. At this stage the project demonstrates a functioning public website, an authenticated student portal, a protected admin panel, a relational database model, tested API behaviour, and documented information architecture and requirement logic.

4.2 Nature of the Current Delivery

The current delivery is technically functional, suitable for demonstration and academic evaluation, and structurally strong as a prototype. It is not yet equivalent to a production-ready enterprise platform; deeper integrations and production-level concerns remain future work. However, the current state already provides a solid proof of concept. This roadmap realistically distinguishes what has been achieved from what remains.

5. Completed Work Summary

5.1 Planning and Analysis

The planning and analysis stage delivered the project proposal definition, scope identification, stakeholder framing, user requirement analysis [3], and information architecture planning. This stage established the strategic and structural logic of the platform.

5.2 Design and Prototyping

Design work delivered the two-level website structure, the route-based page architecture, the separation of public and authenticated layouts, prototype visual refinement for both Level 1 and Level 2, and admin interface design. The Information Architecture Design document [4] records the finalised site map, navigation structure, labelling conventions, taxonomy, and user-flow design.

5.3 Frontend Development

Frontend development produced the homepage and authentication pages, the Level 2 program exploration pages, the research and innovation pages, the Level 1 student portal pages, and the admin dashboard and management pages. The frontend uses Vue 3 with Vue Router, Pinia for state management, and Axios for API communication. All pages are implemented as single-file components with role-aware layouts for each environment.

5.4 Backend Development

Backend development produced the authentication APIs, program and roadmap APIs, search APIs, admin CRUD endpoints, community and discussion endpoints, image upload support, route protection and role control middleware, and error handling middleware. The backend is built with Express on Node.js, uses mysql2 with parameterised queries, and implements JWT-based authentication with role-based authorisation.

5.5 Database and Data Mapping

Database-related work covered schema creation (19 tables), seed data preparation (programs, courses, alumni, industry, users), roadmap mapping logic (program_courses join table), fixes to incorrect course-to-program relationships, and improvements to admin-side program-course linkage during course creation. The schema supports program-roadmap structure, user roles, community features, messaging, and content management.

5.6 Testing and Defect Resolution

Testing work delivered Postman functional testing (10 cases, all passed), Python smoke testing (9 automated checks, all passed), JMeter performance testing (3 scenarios, 1,200 requests, 0% errors), frontend build verification, and documented defect tracking and resolution. A total of 45 defects were identified, tracked, and resolved, with detailed root-cause analysis and fix documentation in the Defect Report [5]. The testing and evaluation results are documented in the System Testing and Evaluation Report [6].

5.7 Documentation

Seven formal project documents were produced covering requirements, information architecture, technical design, implementation planning, testing and defects, and final reporting. All documents are linked through explicit references, forming a coherent project documentation set. The Technical Documentation [7] provides detailed API, database, and component-level design records.

6. Implementation Phases

The project proposal originally described an eight-stage implementation plan: planning, requirement analysis, information architecture, wireframe and UI design, development, testing, deployment, and handover. All eight phases have been substantially addressed in the current prototype, with the deployment phase interpreted as local development and demonstration deployment rather than production deployment. The phased structure follows standard implementation roadmap conventions [8], where each phase builds on the deliverables of the preceding phase.

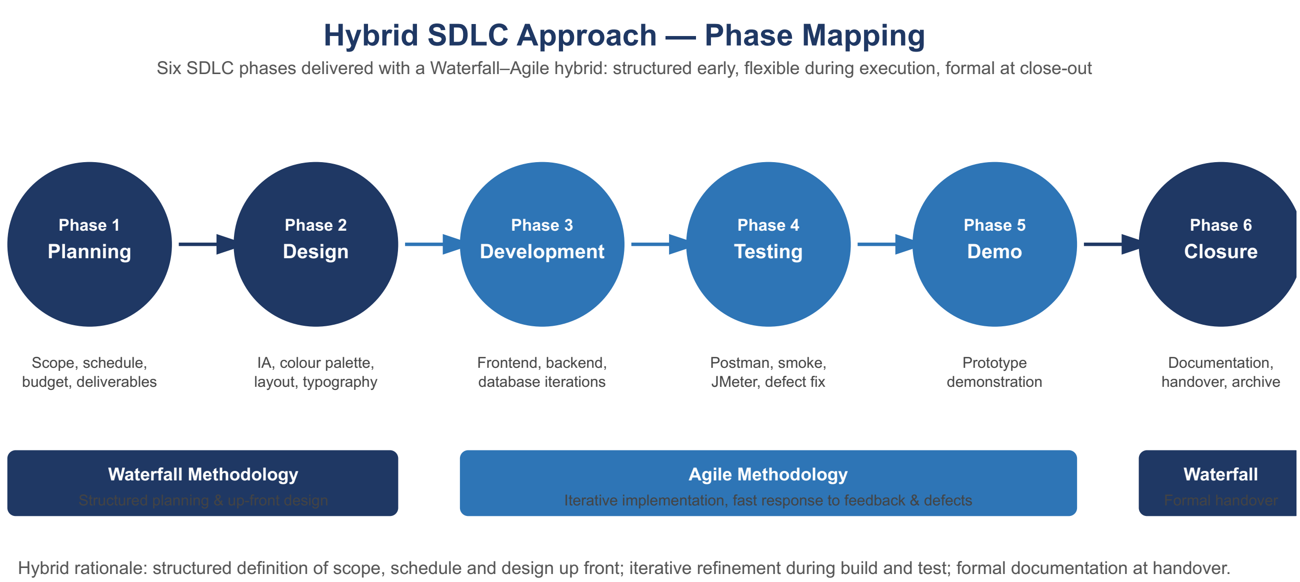

Figure 1. Hybrid SDLC phases.

7. Workstreams and Tasks

The implementation was organised around five parallel workstreams:

Frontend development delivered a Vue 3 single-page application with over 30 pages across four environments.

Backend development produced an Express API with 6 route groups covering authentication, programs, search, admin CRUD, community, and profile operations.

Database design and maintenance produced a normalised MySQL schema with 19 tables, seed data, and roadmap mapping between programs and courses.

Testing and quality assurance covered Postman, Python smoke tests, JMeter, and build verification.

Documentation produced 7 formal documents covering requirements through to final reporting, as recommended by PMI planning practice [2].

8. Timeline Plan

The project timeline spanned approximately 10 weeks from initial proposal to final documentation. Weeks 1–2 focused on planning and requirement analysis. Weeks 3–4 covered information architecture and initial design. Weeks 5–7 covered core development (frontend and backend). Weeks 8–9 covered testing, defect resolution, and documentation refinement. Week 10 focused on final packaging, presentation preparation, and formal handover.

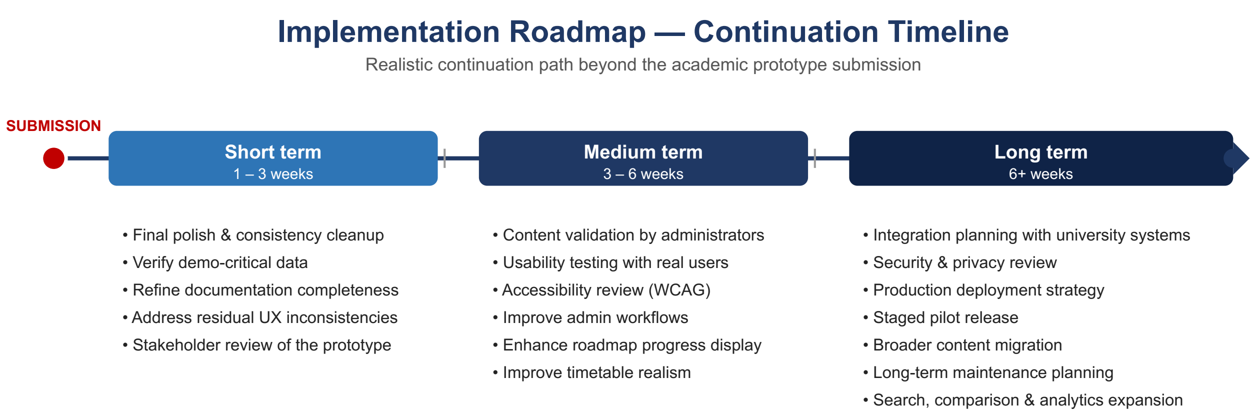

Figure 2. Implementation Roadmap timeline.

9. Key Milestones

Milestone	Description
Project Proposal	Completed Week 2: scope, objectives, and initial plan defined [1]
Requirement	Completed Week 3: user requirement analysis finalised [3]

Analysis	
IA Design	Completed Week 4: information architecture design completed [4]
Core Implementation	Completed Week 7: frontend and backend development substantially complete
Testing & Defects	Completed Week 9: all 45 tracked defects resolved; Postman, smoke, and JMeter tests all passing at 0% errors [5, 6]
Final Documentation	Completed Week 10: all 7 formal documents completed and packaged
Presentation & Handover	Week 10: final presentation, demo, and handover

Table 2. Key project milestones.

10. Roles and Responsibility Alignment

The project team operated with shared responsibilities across the implementation workstreams. Frontend development was primarily led by J. Liu with support from the full team. Backend development was shared across team members, with J. Liu leading API architecture and integration. Testing and quality assurance was coordinated by J. Liu, with all team members contributing to defect identification and verification. Documentation was produced collaboratively, with each document assigned a primary author and reviewed by the full team before finalisation.

11. Dependencies

11.1 Technical Dependencies

The frontend depends on the backend for all data operations via REST APIs. The backend depends on MySQL for persistent storage and on the .env configuration for database credentials, JWT secret, and Cloudinary settings. The testing layer depends on a running backend instance, valid seeded data, consistent API response formats, and predictable authentication behaviour.

11.2 Data Dependencies

The quality of the roadmap and public content depends on accurate seeded data, correct program-course mappings, and valid links between programs and related content domains such as alumni, industry, and career outcomes.

11.3 Documentation Dependencies

Later deliverables depend on earlier deliverables: the Final Report depends on the requirement report, the IA design, the testing record, and the technical documentation; the presentation depends on the final report and demo script; and this implementation roadmap depends on the actual implemented system state.

12. Risks and Contingency Considerations

12.1 Technical Risks

Risk	Impact	Mitigation Direction
Reliance on seeded data	Reduces realism of some modules	Verify and refine critical records before demo
Timetable remains non-live	Limits practical realism	Document limitation clearly; define future integration path
Local upload model is not production-grade	Limits deployment readiness	Redesign storage approach in a future production phase
Expanding too many modules at once	Quality dilution	Keep roadmap-centred priorities clear

Table 3. Technical risk assessment.

12.2 Project Risks

Risk	Impact	Mitigation Direction
Documentation written too late	Weak final handover	Complete major project documents before final packaging
Scope creep into full enterprise solution	Reduced quality and feasibility	Keep project framed as prototype with planned extension path
Presentation drifts from roadmap concept	Weaker narrative coherence	Keep roadmap as central theme in reporting and demo

Table 4. Project risk assessment.

12.3 Contingency Logic

The current project reduces future risk because the architecture is modular, major defects have already been addressed, core features have already been tested, and the remaining work is enhancement-oriented rather than foundational. Future iterations can proceed without needing to rebuild structural components.

13. Feasibility Assessment

The feasibility of continuing the project rests on concrete evidence from the current prototype. The frontend, backend, database, and testing layers are all functioning together in a coherent system. The largest architectural decisions—the two-level design, role-based access, roadmap-centred data model, and modular structure—have already been made, implemented, and tested.

Specific evidence includes: 30+ implemented frontend pages across four environments; six functioning API route groups with consistent response handling; a 19-table normalised database with tested seed data; 10 Postman test cases, 9 smoke tests, and 3 JMeter scenarios all passing at 0% error rates; and 45 defects resolved with root-cause analysis. The project has moved past conceptual uncertainty into a stage where continued value comes from controlled refinement rather than foundational reconstruction.

14. Recommended Next Steps

14.1 Immediate Next Steps

1. Finalise all formal deliverables and presentation assets. 2. Verify demo-critical flows and seeded records one final time. 3. Strengthen the link between dashboard, roadmap, and timetable where possible. 4. Improve admin workflows for future maintainability. 5. Prepare a realistic production-oriented recommendation path rather than attempting immediate production deployment.

14.2 Longer-Term Steps

1. Timetable integration with real institutional scheduling data. 2. Deeper roadmap progress analytics. 3. Public program comparison tools. 4. Broader accessibility validation and automated UI testing. 5. Production deployment planning, including hosting, environment hardening, and CI/CD.

15. Conclusion

The Adelaide University Program Roadmap project has reached a strong prototype milestone. The implementation roadmap shows a clear progression from planning to design, development, testing, and documentation, and demonstrates that the remaining work is manageable rather than foundational.

The project is feasible as a working structured system, not merely as a concept. The current prototype proves the technical viability of the roadmap-centred, role-aware approach. The next stage is a disciplined continuation that strengthens realism, polish, and deployment readiness while preserving the user focus established in the current system.

References

[1] J. Liu, Y. Huang, Y. Peng, and V. Thanki, "Development of Program Roadmap — Project Proposal," Group C261W-4037, Industry Research Project, Adelaide University, Mar. 18, 2026.

[2] Project Management Institute, A Guide to the Project Management Body of Knowledge (PMBOK Guide), 7th ed. Newtown Square, PA, USA: PMI, 2021.

[3] J. Liu, Y. Huang, Y. Peng, and V. Thanki, "AU Roadmap User Requirement Analysis Report," Group C261W-4037, Industry Research Project, Adelaide University, May 10, 2026.

[4] J. Liu, Y. Huang, Y. Peng, and V. Thanki, "AU Roadmap Information Architecture Design," Group C261W-4037, Industry Research Project, Adelaide University, May 10, 2026.

[5] J. Liu, Y. Huang, Y. Peng, and V. Thanki, "AU Roadmap Defect Report," Group C261W-4037, Industry Research Project, Adelaide University, 2026.

[6] J. Liu, Y. Huang, Y. Peng, and V. Thanki, "AU Roadmap System Testing and Evaluation Report," Group C261W-4037, Industry Research Project, Adelaide University, May 9, 2026.

[7] J. Liu, Y. Huang, Y. Peng, and V. Thanki, "AU Roadmap Technical Documentation," Group C261W-4037, Industry Research Project, Adelaide University, May 10, 2026.

[8] ProjectManager, "Implementation Roadmap." [Online]. Available: https://www.projectmanager.com/blog/implementation-roadmap